PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Contrast extravasation and outcome of endovascular therapy in acute ischemic stroke: a systematic review and meta-analysis
AUTHORS	Xu, Tao; Wang, You; Yuan, Jinxian; Chen, Yang-Mei; Luo, Haiyan

VERSION 1 – REVIEW

REVIEWER	Zhivolupov, S. A. Mil Med Acad, neurology
REVIEW RETURNED	18-Oct-2020

GENERAL COMMENTS	A high-quality translation of the text into English shoud be done, the are a lot of the stylistic mistakes.

REVIEWER	Barboza, Miguel Hospital Dr. Rafael A. Calderón Guardia, CCSS, Neurosciences Department
REVIEW RETURNED	30-Oct-2020

GENERAL COMMENTS	I had the pleasure the read the manuscript "Contrast extravasation and outcome of endovascular therapy in acute ischemic stroke: a systematic review and meta-analysis", which presents the strenght of association between CE and unfavourable functional outcomes and intracranial haemorrhage in acute stroke cases undergoing mechanical thrombectomy. The analysis is robust, well written and complies with MOOSE criteria for meta-analysis. According to the editor's request, I'll give my comments based on the statistical analysis. a) The random-effects model was used by the authors, but clear justification should be stated in this section, in terms on how the model accounts for predictors of CE in functional and hemorrhagic outcomes b) In terms of publication bias, did you perform a funnel plot analysis (together with funnel plot asymmetry)? If so, I consider this information relevant, and the figure of this analysis could also be included as a supplement c) In figure 2,3 and 4, forest plots should improve showing the number of CE + cases vs. non-CE cases, for further reproducibility of the analysis. d) In figure 2 and 3, to improve the readers' experience, data analysis at discharge, or in-hospital, should be placed first, and after that, data related to 90-day assessments. e) In all forest plot figures, please add legends related to the association of the result (i.e. "favours mortality risk")

REVIEWER	Freeze, Whitney

	Leiden Universitair Medisch Centrum
REVIEW RETURNED	12-Jan-2021

GENERAL COMMENTS	The manuscript constitutes a systematic review and meta-analysis on the association between CT contrast extravasation and outcome variables (poor functional outcome (modified ranking scale score at least 3) at 90 days, poor functional outcome at discharge, mortality at 90 days, in-hospital mortality, intracranial haemorrhage and symptomatic ICH) after endovascular therapy in acute ischemic stroke. The topic is relevant and results are interesting. However, I have several suggestions for improvement: - The authors should include the definition of CE in the inclusion criteria - The authors should include outcome definitions in the inclusion criteria. How were intracranial hemorrhage and symptomatic ICH after EVT assessed and defined? - CE was defined as absence of a hypointensity on T2*-weighted MRI 24 hours after EVT, does this mean that cases with hypointensity on T2*-weighted MR images at 24 hours and high density on NECT immediately after EVT were considered to be CE negative but ICH positive? - The authors should state exclusion criteria in the text (figure 1 shows reasons for exclusion; what is meant by 'insufficient data' and 'not relevant outcomes'?). - What type of contrast material was administered in each study during EVT? - The literature search was conducted by two independent authors; how many disagreements about the inclusion of a study did the authors have? - How many authors performed the data extraction? Ideally, this should be done by at least 2 authors, with consensus agreement on the extracted data. - Did the authors check whether there was any overlap between studies with regard to the included study participants? - Does the study period in table 2 refer to the years during which participants were included for each study? Please clarify. - Can the authors indicate the number of participants with/without CE in each cohort? - Page 9, line 51-60 & page 10, line 4-15; can the authors, apart from the number of studies, also indicate how many individuals were included in the meta-analysis in the text? - The I squared statistic can be biased when the number of studies is small (see https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/s12874-015-0024-z) , can the authors provide the confidence interval? - There is probably a typo in the conclusion on page 12, line 48/49: '… patients with AIS undergoing MT'. MT is not defined in the text.

VERSION 1 – AUTHOR RESPONSE

Reviewer Reports:

Replies to Reviewer #1

1. "A high-quality translation of the text into English should be done, there are a lot of the stylistic mistakes"

Response: We are sorry for the stylistic mistakes. In this revision, we have rewritten our manuscript and also revised stylistic mistakes. Moreover, this revised manuscript has been edited and proofread by American Journal Experts.

Replies to Reviewer #2

1. “The random-effects model was used by the authors, but clear justification should be stated in this section, in terms on how the model accounts for predictors of CE in functional and hemorrhagic outcomes”

Response: In this revision, we have added this information in the Methods section (Statistical analysis). In this meta-analysis, we recognized the potential heterogeneity between the included studies; thus, we used a random effects model to pool the estimates for improving the reliability of the results. (Methods section, page 9)

2. “In terms of publication bias, did you perform a funnel plot analysis (together with funnel plot asymmetry)? If so, I consider this information relevant, and the figure of this analysis could also be included as a supplement”

Response: In this revision, we performed funnel plot analyses. We identified asymmetric funnel plots involved in the pooled estimates (figure S1-S4 in the online data supplement). However, Egger’s tests showed no publication bias in the pooled estimates (table S5 in the online data supplement).

3. “In figure 2, 3 and 4, forest plots should improve showing the number of CE + cases vs. non-CE cases, for further reproducibility of the analysis.”

Response: We extracted the ORs and 95% CIs or raw data to calculate ORs for the association between CE and the outcomes of EVT. We extracted the OR from the most fully adjusted model (showed in the figure 2-4). The raw data (the number of CE + cases vs. non-CE cases) to calculate ORs for the association between CE and the outcomes of EVT has been showed in the table S2 in the online data supplement, which could be used for reproducibility of the analysis.

4. “In figure 2 and 3, to improve the readers' experience, data analysis at discharge, or in-hospital, should be placed first, and after that, data related to 90-day assessments.”

Response: We have revised this inappropriate description and rearranged the order of these results. (Results section, page 10-11, figure 2-3)

5. “In all forest plot figures, please add legends related to the association of the result (i.e. "favours mortality risk")”

Response: We have added this information in the figure 2-4 and their figure legends (favours increased and decreased risk).

Replies to Reviewer #3

1. “The authors should include the definition of CE in the inclusion criteria”

Response: We have added the definition of CE in the inclusion criteria. (Methods section, page 6, table 1)

2. “The authors should include outcome definitions in the inclusion criteria. How were intracranial hemorrhage and symptomatic ICH after EVT assessed and defined?”

Response: We have added outcome definitions in the inclusion criteria. We have added more information regarding the definitions of ICH and sICH. ICH after EVT was defined as any hemorrhage, including hemorrhagic infarction, parenchymal hemorrhage, or intracranial-extracerebral hemorrhage (Stroke 2015; 46:2981-6.), which was detected with CT or MRI scan after CE assessment. sICH was defined as post-EVT ICH with significant neurological deterioration (National Institutes of Health

Stroke Scale (NIHSS) scores ≥ 4 in total) (Methods section, page 7). Moreover, the assessment strategies of hemorrhage after EVT have been demonstrated in the Table 1.

3. “CE was defined as absence of a hypointensity on T2*-weighted MRI 24 hours after EVT, does this mean that cases with hypointensity on T2*-weighted MR images at 24 hours and high density on NECT immediately after EVT were considered to be CE negative but ICH positive?”
Response: Based on the included studies and our clinical practice, CE was defined as an absence of a hypointensity on 24 hours follow-up MRI-GRE and MRI-SWI after EVT (not T2-weighted MRI); ICH was defined as a presence of a hypointensity on 24 hours follow-up MRI-GRE and MRI-SWI after EVT (The assessment strategies of contrast extravasation and ICH after EVT are summarized in the table 1).

4. “The authors should state exclusion criteria in the text (figure 1 shows reasons for exclusion; what is meant by ‘insufficient data’ and ‘not relevant outcomes”.
Response: We have added the exclusion criteria in the Methods section (Methods section, page 7). If one study did not explore the associations between CE and the outcomes of EVT, it was considered as a study with irrelevant outcomes (cannot meet the first in the inclusion criteria). One study without any data regarding outcome assessment after EVT was considered as insufficient data (cannot meet the fifth in the above inclusion criteria)

5. “What type of contrast material was administered in each study during EVT?”.
Response: Iodine-based contrast media was commonly administered in those included studies. Most of the included studies did not provide the detailed information of contrast material (e.g. name, dose, and pharmaceutical company).

6. “The literature search was conducted by two independent authors; how many disagreements about the inclusion of a study did the authors have?”
Response: There are two disagreements about the inclusion of a study:
1) First disagreement: A study: World neurosurgery 2019, 130:e583-e587 and B study: Aging and disease 2019, 10(4):784-792. One author considered that the two studies have a potentially significant sample overlap due to the overlap of inclusion period of patients in a same place. However, after we checked the two articles carefully, we found that the A study mainly included patients with anterior cerebral circulation occlusion; the B study included patients with posterior cerebral circulation occlusion. Thus, the two studies have different study cohort, and both of them should be included in this meta-analysis.
2) Second disagreement: C study: Journal of neurology 2012, 259(5):936-943 and D study: Stroke 2004, 35(4):876-881. One author considered that the two studies have a potentially significant sample overlap due to that the two studies were conducted in a same institute. However, after we discussed this disagreement, we found that the two studies have different inclusion period of participants. Thus, we considered that there is no significant sample overlap between the two studies, and both of them should be included in this meta-analysis.

7. “How many authors performed the data extraction? Ideally, this should be done by at least 2 authors, with consensus agreement on the extracted data.”
Response: Two reviewers independently extracted the relative data from each article with consensus agreement on the extracted data (Methods section, page 8).

8. “Did the authors check whether there was any overlap between studies with regard to the included study participants?”
Response: We have checked this in the study selection. In this meta-analysis, if there was a potentially significant sample overlap among multiple studies, the study with the longest follow-up or highest sample size was included (Methods section, page 8).

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9. “Does the study period in table 2 refer to the years during which participants were included for each study? Please clarify.”
Response: The study period of each study in table 2 is the inclusion period of participants. We have revised this in the table 2.

10. “Can the authors indicate the number of participants with/without CE in each cohort?”
Response: The information has been showed in the table S2 in the online data supplement. We extracted the ORs and 95% CIs or raw data to calculate ORs for the association between CE and the outcomes of EVT. We extracted the OR from the most fully adjusted model (showed in the figure 2-4). The raw data (the number of participants with/without CE) to calculate ORs for the association between CE and the outcomes of EVT has been showed in the table S2 in the online data supplement, which could be used for reproducibility of the analysis.

11. “Page 9, line 51-60 & page 10, line 4-15; can the authors, apart from the number of studies, also indicate how many individuals were included in the meta-analysis in the text?”
Response: We have added this information in the result section and abstract section. We have revised this inappropriate description and rearranged the order of these results. (Results section, page 10-11)

12. “The I squared statistic can be biased when the number of studies is small (see https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/s12874-015-0024-z) , can the authors provide the confidence interval?”
Response: Based on your suggestion, we also estimated 95% CIs to assess the heterogeneity between estimates (Methods section, page 9). For heterogeneity assessments of pooled estimates, the I2 and 95% CIs were listed in the figure legends.

13. “There is probably a typo in the conclusion on page 12, line 48/49: ‘… patients with AIS undergoing MT’. MT is not defined in the text”.
Response: We are sorry for this mistake. We have revised it.

VERSION 2 – REVIEW

REVIEWER	Barboza, Miguel Hospital Dr. Rafael A. Calderón Guardia, CCSS, Neurosciences Department
REVIEW RETURNED	16-Mar-2021

GENERAL COMMENTS	The authors addressed all my suggestions.